

MARKETING, BIG LOGIC AND A PINCH OF ACID (without *ACID*)

ABOUT ME

Alejandro M. Ramallo - Founder/CEO [Leapsight](#)

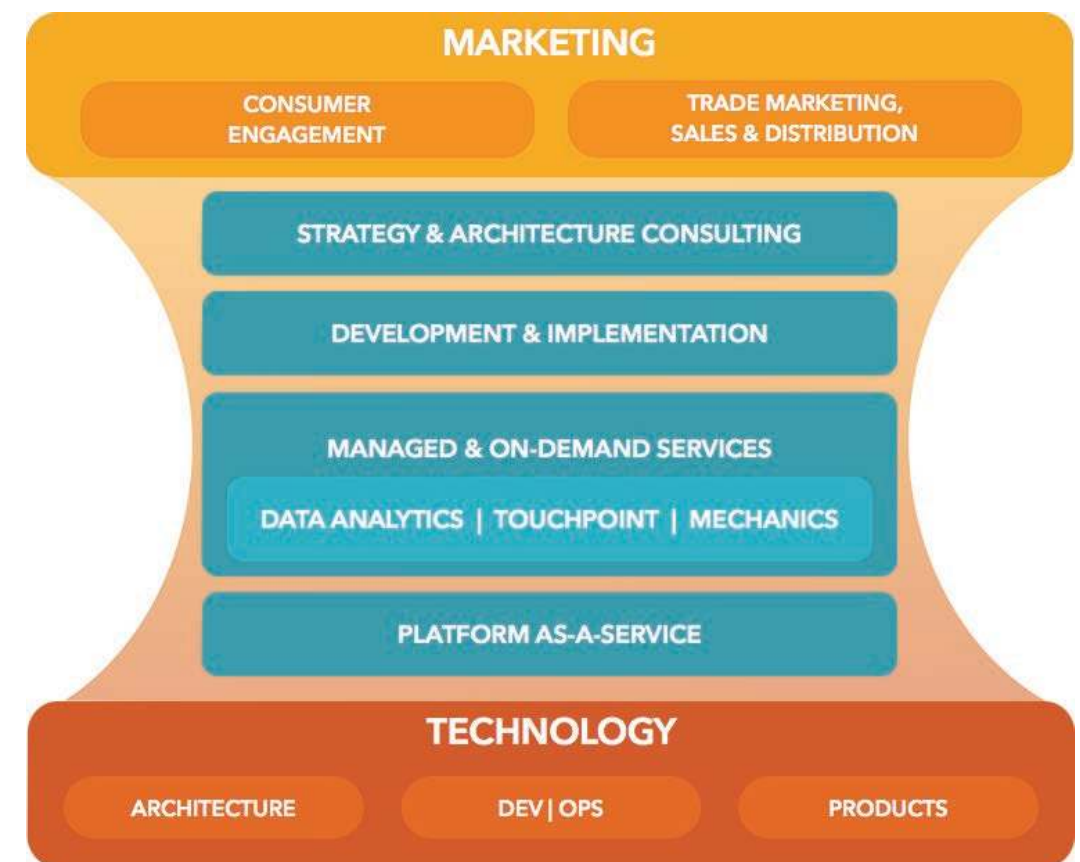
- Born in Argentina, married to an artist and father of 3 beautiful children
- Living / working in UK since 2006
- MSc. equiv. Business degree
- Former Group HoT, Group HoMIT, Mercosur CIO & Consumer Engagement Director @ FMCG

ABOUT ME (cont.)

- Architecting marketing & sales automation technology solutions since 1997
- Objective-C OpenStep/WebObjects, Java/J2EE/JINI/Javaspace, Erlang/OTP
- Business Intelligence & Data Analytics, Rule-based/Expert Systems



- Started operations in 2011
- Team with strong background in architecture and implementation of bespoke scalable, high availability, high performance Consumer Marketing Front Office solutions
- Headquartered in London, U.K. with R&D in Buenos Aires and Rosario, Argentina
- Investing for 3 years in the development of technologies and frameworks to radically change the way consumer marketers do marketing in the digital age







PUSH MASS MARKETING

low-res :: 1 segment



PUSH SEGMENTATION | CRM

medium-res :: a handful of segments



HIGHT EXPECTATIONS

- Permission-based (~~PUSH~~)
- Consistent omni-touchpoint experience
- Customisation
- Entertainment & Service
- Participation | Social Contribution
- Always-on & Real-time



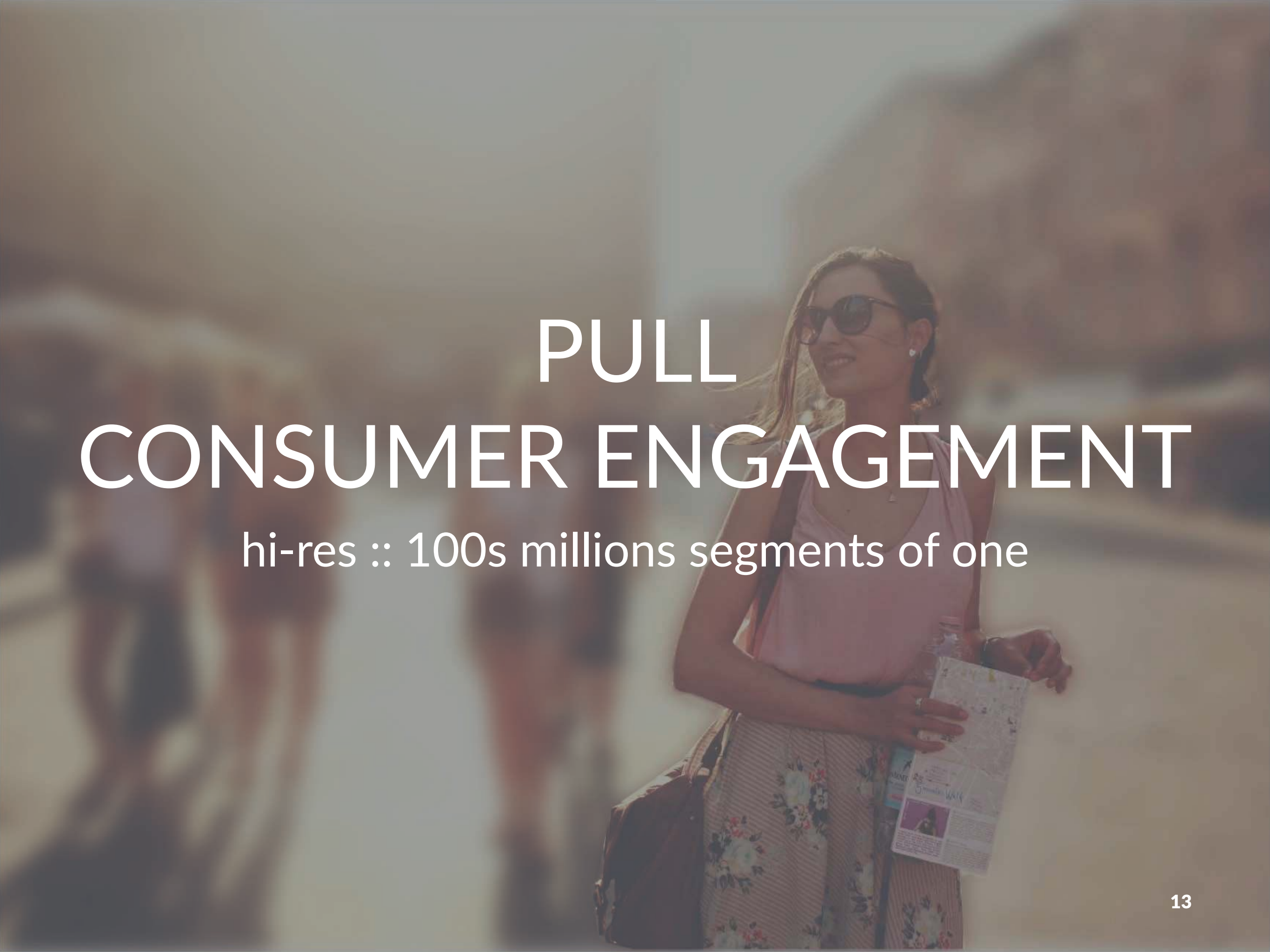


A woman with blonde hair and sunglasses is walking on a cobblestone street in a city. She is wearing a pink tank top and a floral skirt, and is holding a map. The background shows other pedestrians and historic buildings. Overlaid on the image are several dark grey circles containing the words 'Sports', 'Travel', 'Gaming', 'Read', and 'Travel'. There are also several teal circles containing social media icons: Facebook (f), Twitter (bird), Google Plus (g+), and LinkedIn (in). A red location pin icon is also visible on the right side of the image.

Consumers are more powerful than ever before



Word-of-mouth has become
the most trusted source of
advertising

A woman with long blonde hair, wearing sunglasses, a pink tank top, and a floral skirt, is smiling and looking to her right. She is holding a map and a water bottle. The background is a blurred crowd of people on a beach, suggesting a sunny, outdoor setting.

PULL CONSUMER ENGAGEMENT

hi-res :: 100s millions segments of one

THE WEB IS THE DATABASE

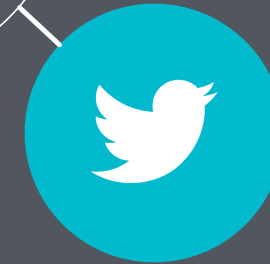
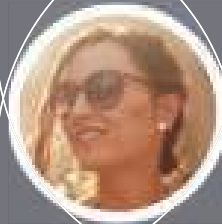




```
first_name: Mary
middle_name: Jane
surname: Farrell
dob: 24/11/99
married: false
friends: [...]
regular_brand: X
daily_consumption: 2
modified: 1m ago
```

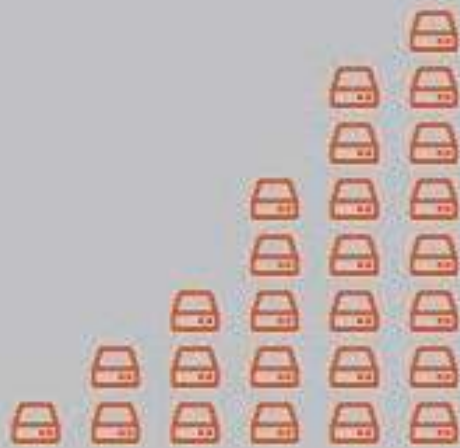


```
name: Mary Farrell
location: NY
follows: [...]
modified: 1w ago
```



THUS, BIG DATA

VOLUME



Store, manage and process data.

VELOCITY

Every 60 seconds

 100k

 700k

 11m

Transform into actionable knowledge.

VARIETY



Hundreds of sources.
Different formats.

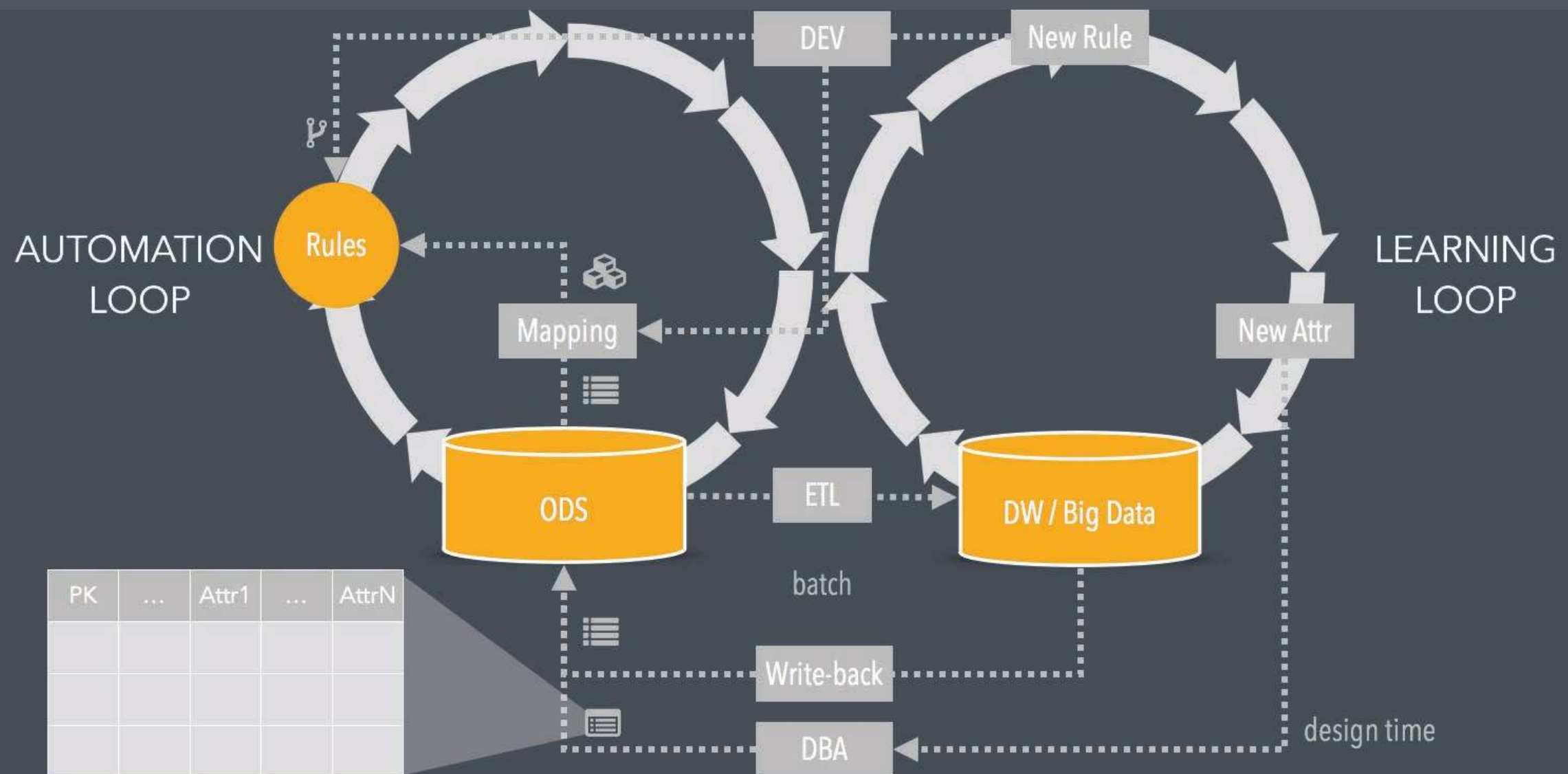
DATA ITSELF IS NOT AN ASSET,
ACTIONABLE KNOWLEDGE IS

A proposition is actionable
when actors **can** use it to
effectively implement their
intentions

— *Chris Argyris - The Oxford Handbook of Organization
Theory*

INTELLIGENT
DATA ANALYSIS
IS THE QUEST FOR
ACTIONABLE KNOWLEDGE

MAINSTREAM ARCHITECTURE



ACTIONABLE

IS AN **EXTRINSIC** QUALITY ATTRIBUTE

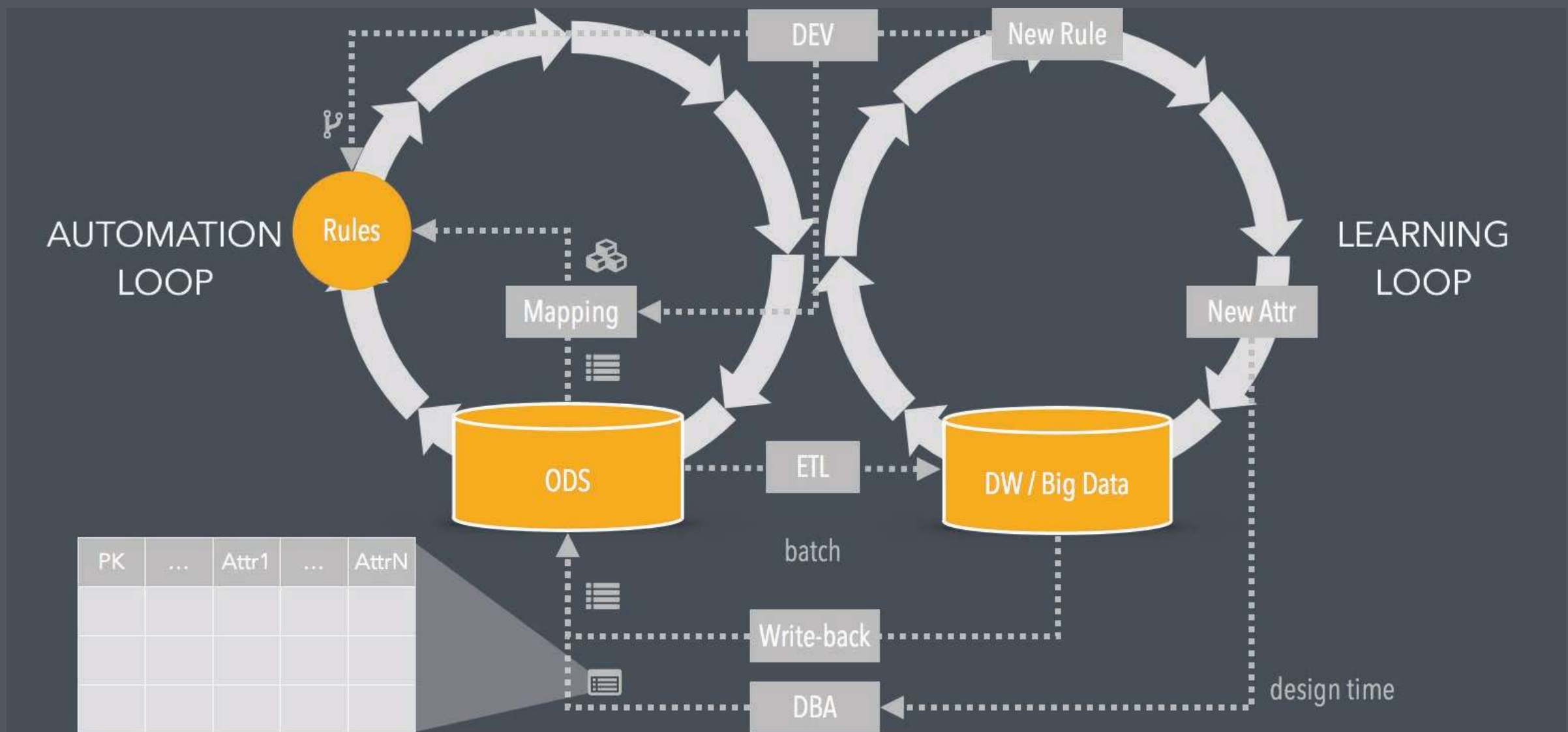
IN ORDER TO EXECUTE WE NEED

- Scalability
- Consistency (omnichannel brand experience)
- Always-on (continuous availability R|W)
- Real-time (decisioning)
- Big, Highly-connected Data
- **Plasticity** (being easily moulded)
- **Agility** (move quickly)

A red vase is shown in the process of shattering on a light-colored surface. The vase is broken into several pieces, with many small red fragments scattered around it. The background is a soft, out-of-focus grey.

TRADITIONAL MARKETING SOLUTIONS
LACK THE CAPABILITY
(PLASTICITY AND AGILITY)
TO EXECUTE
CONSUMER ENGAGEMENT

WE NEED TO REDUCE COMPLEXITY





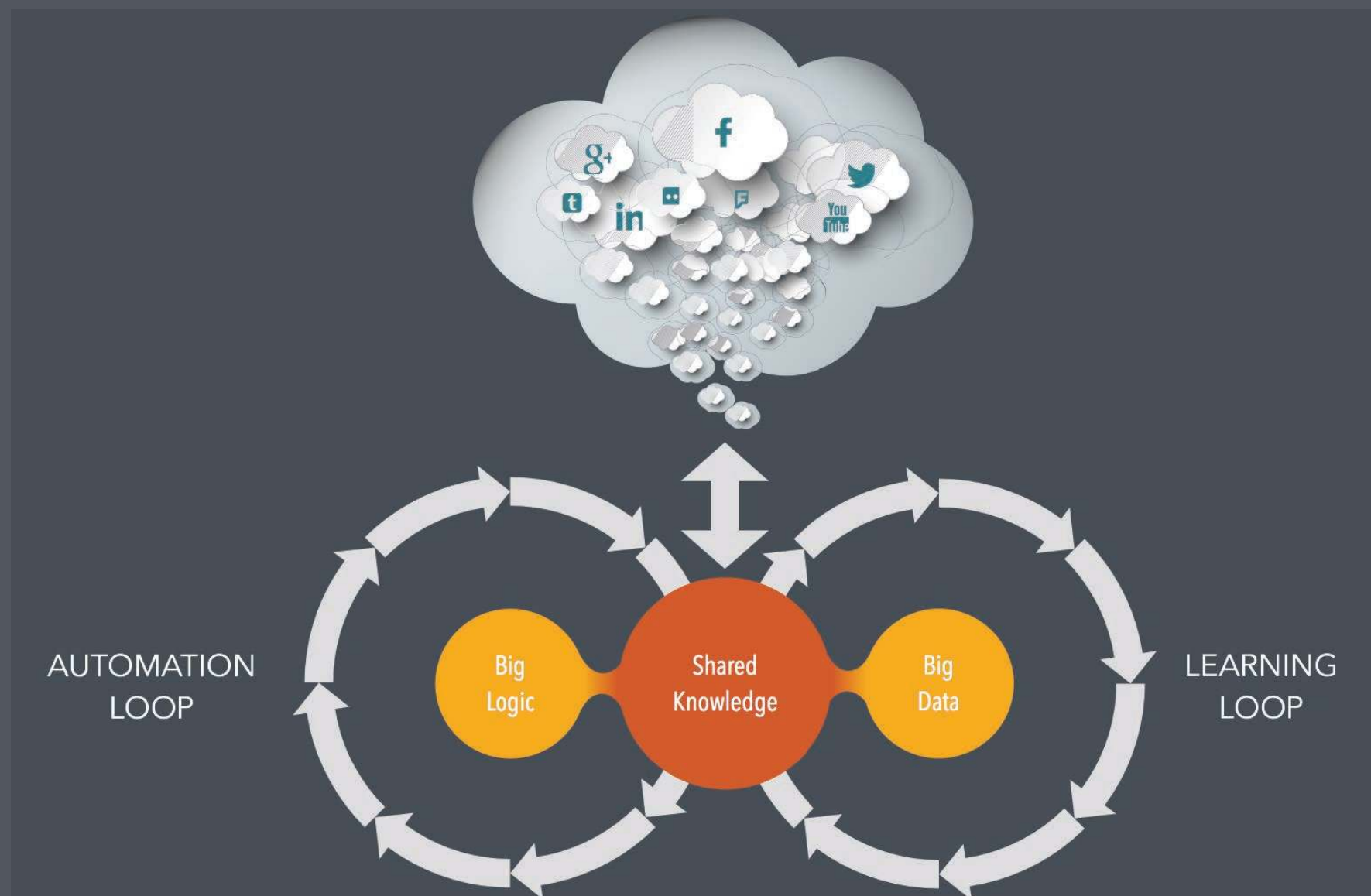
BIG DATA

+

BIG LOGIC

CONTINUUM ARCHITECTURE

THAT MINIMISES ACCIDENTAL COMPLEXITY



INFORMATION MODEL REQUIREMENTS

- FACTS
- FLAT
- UNIFORM
- UNIVERSAL

RELATIONAL, COMPOSABLE VIA FUNCTIONS

INTO INTENSIONAL|DERIVED RELATIONS OR VIEWS

Functional Relational Programming [Our of the Tar Pit](#), B. Moseley, P. Marks

~~SQL TABLES~~

~~{KEY, VALUE}~~

~~DOCUMENT~~

~~OBJECTS~~

~~OO GRAPHS~~

EAV? -> RDF?

EAV

ENTITY	ATTRIBUTE	VALUE
123	:firstName	"Mary"
123	:married	TRUE
123	:dob	24/01/1992

~~EAV~~ RDF TRIPLE

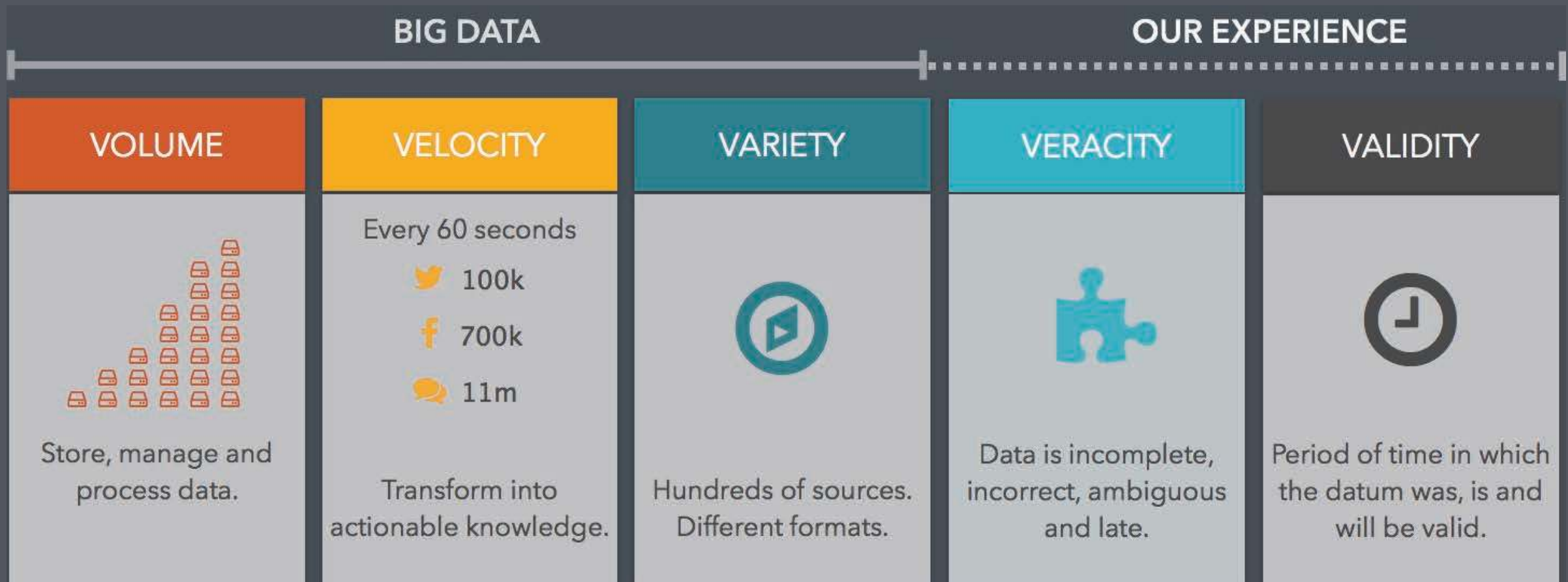
SUBJECT	PREDICATE	OBJECT
<id:123>	:firstName	"Mary"
<id:123>	:married	"true"^^xsd:boolean
<id:123>	:dob	"24/01/1992"^^xsd:date

RDF ~~TRIPLE~~ QUAD

SUBJECT	PREDICATE	OBJECT	GRAPH
<id:123>	:firstName	"Mary"	<Lsd:consumer:profile>
<id:123>	:married	TRUE	<Lsd:consumer:profile>
<id:123>	:dob	"24/01/1992"^^xsd:date	<Lsd:consumer:profile>
<id:123>	fb:friend	<id:524>	<Lsd:consumer:facebook>
<id:123>	fb:firstName	"Kate"	<Lsd:consumer:facebook>
<id:123>	fb:relationship_status	"Single"	<Lsd:consumer:facebook>

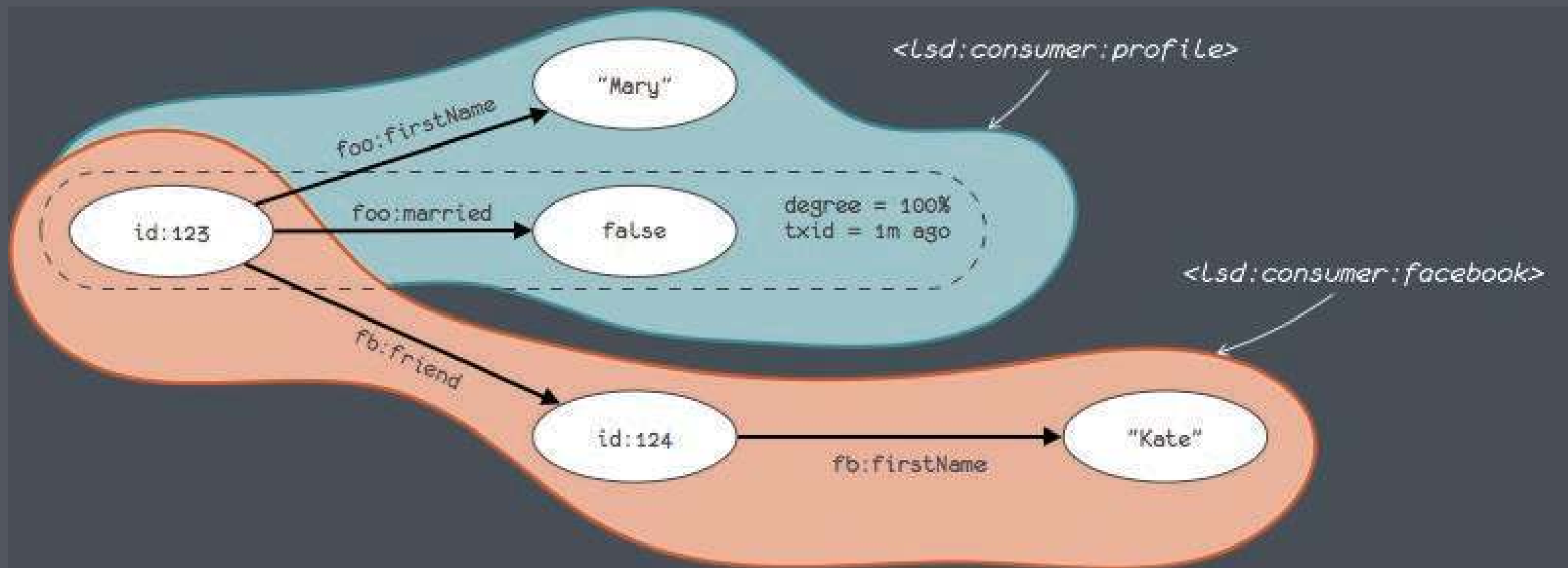
FACTS ARE IMMUTABLE,
SO OUR RECORDS REPRESENTING FACTS
SHOULD BE IMMUTABLE TOO!

TWO MORE THINGS...



STATEMENT

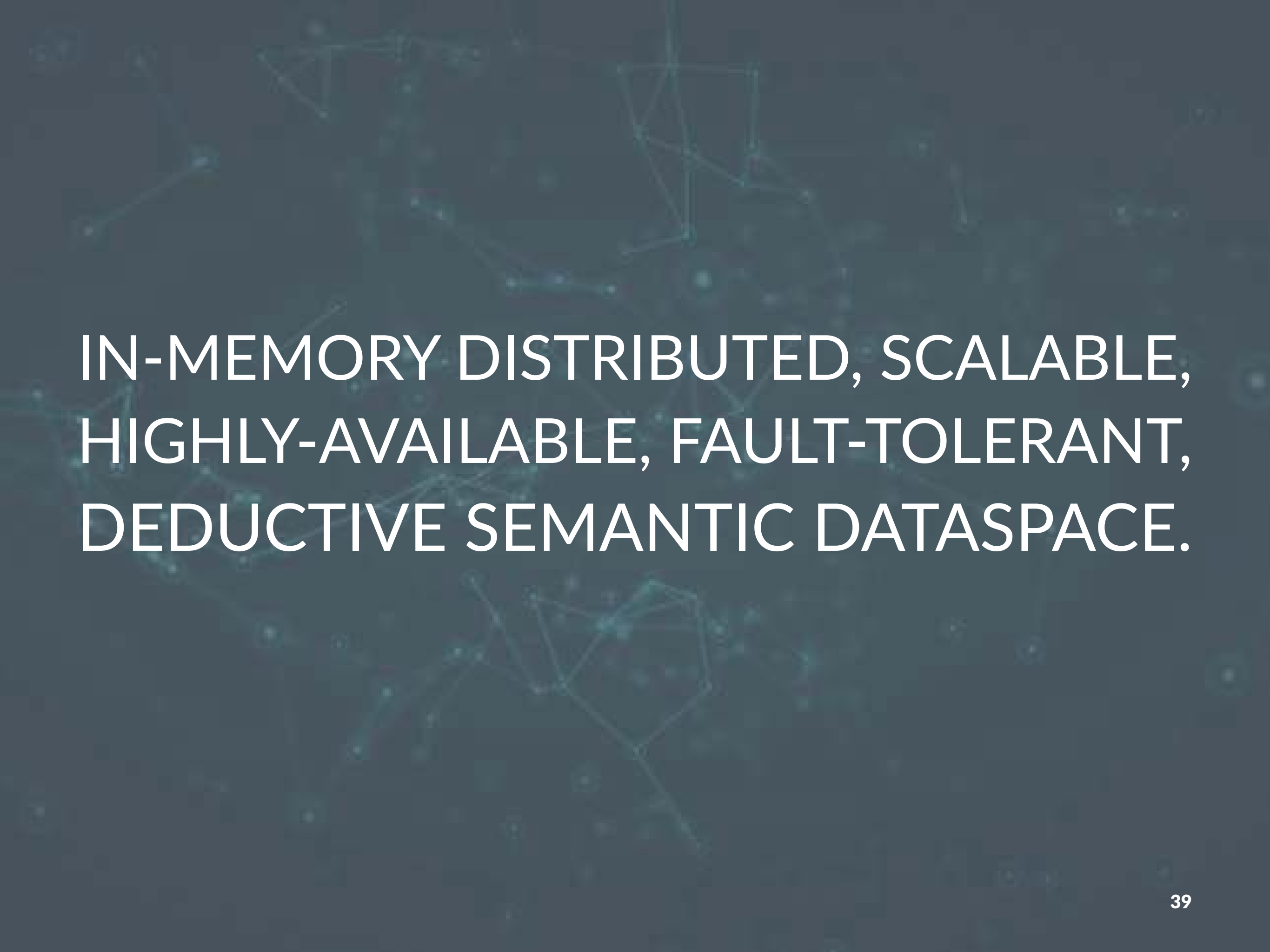
SUBJECT	PREDICATE	OBJECT	GRAPH	DEGREE	TXID
<id:123>	:firstName	"Mary"	<lsd:consumer:profile>	100%	1w ago
<id:123>	:married	TRUE	<lsd:consumer:profile>	100%	1w ago
<id:123>	:dob	"24/01/1992"^^xsd:date	<lsd:consumer:profile>	100%	1w ago
<id:123>	fb:friend	<id:524>	<lsd:consumer:facebook>	100%	1d ago
<id:123>	fb:firstName	"Kate"	<lsd:consumer:facebook>	100%	1d ago
<id:123>	fb:relationship_status	"Single"	<lsd:consumer:facebook>	100%	1d ago
<id:123>	fb:relationship_status	"Single"	<lsd:consumer:facebook>	60%	just now



LEAPSIGHT SEMANTIC DATASPACE

LSID

MARKETING ON ACID



IN-MEMORY DISTRIBUTED, SCALABLE,
HIGHLY-AVAILABLE, FAULT-TOLERANT,
DEDUCTIVE SEMANTIC DATASPACE.

LSD

- A database of *immutable*, *temporal*¹ and *probabilistic* facts called **Statements**

```
-record(lsd_statement, {  
    subject          :: lsd_resource_value(),  
    predicate        :: lsd_uri(),  
    object           :: lsd_value(),  
    graph            :: lsd_uri(),  
    txid             :: binary(),  
    degree = 1.0     :: float(),  
    type             :: lsd_statement_type()  
}).
```

¹ txid, a 128-bit conflict-free k-ordered identifier - github.com/boundary/flake

LSD...BUT NO ACID

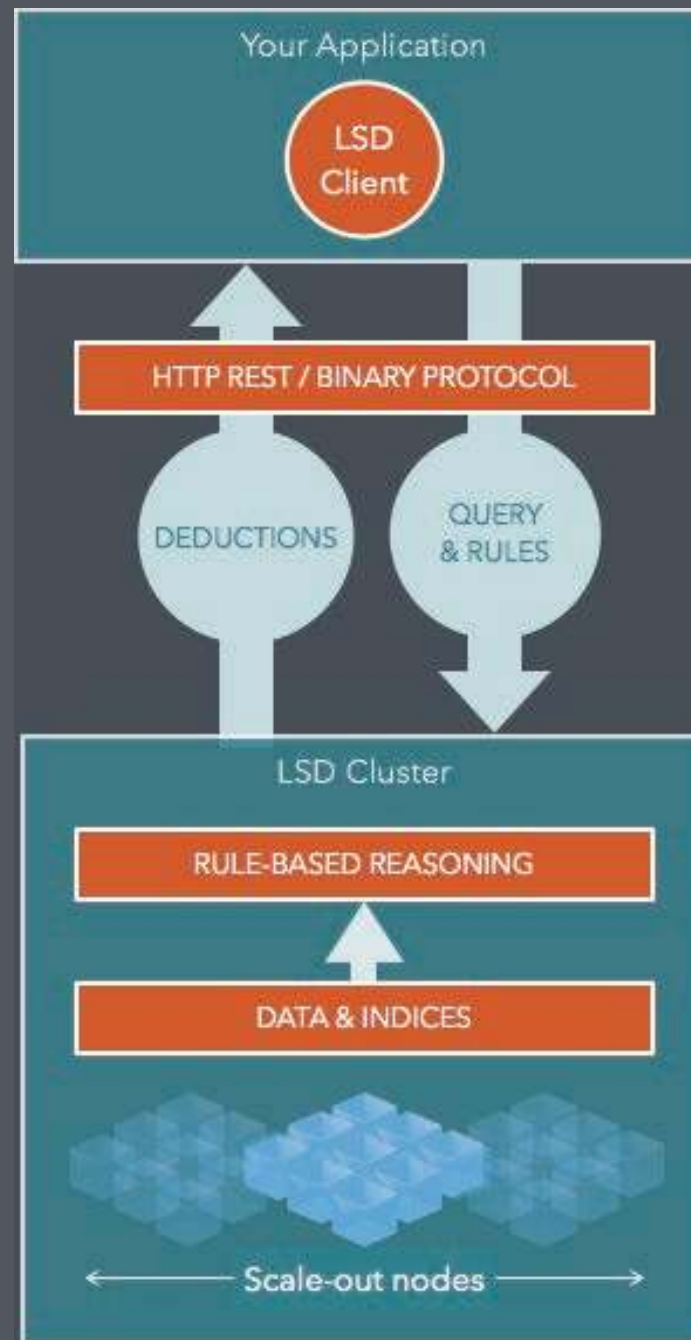
- Eventually consistent
- Append-only store model approximates³ commutative replicated set
- Relational universal data model (NoSQL)
- Stores statements : probabilistic, temporal facts
- Declarative Query language based on Datalog (high level queries)

³ corner cases require more work, demonstration pending

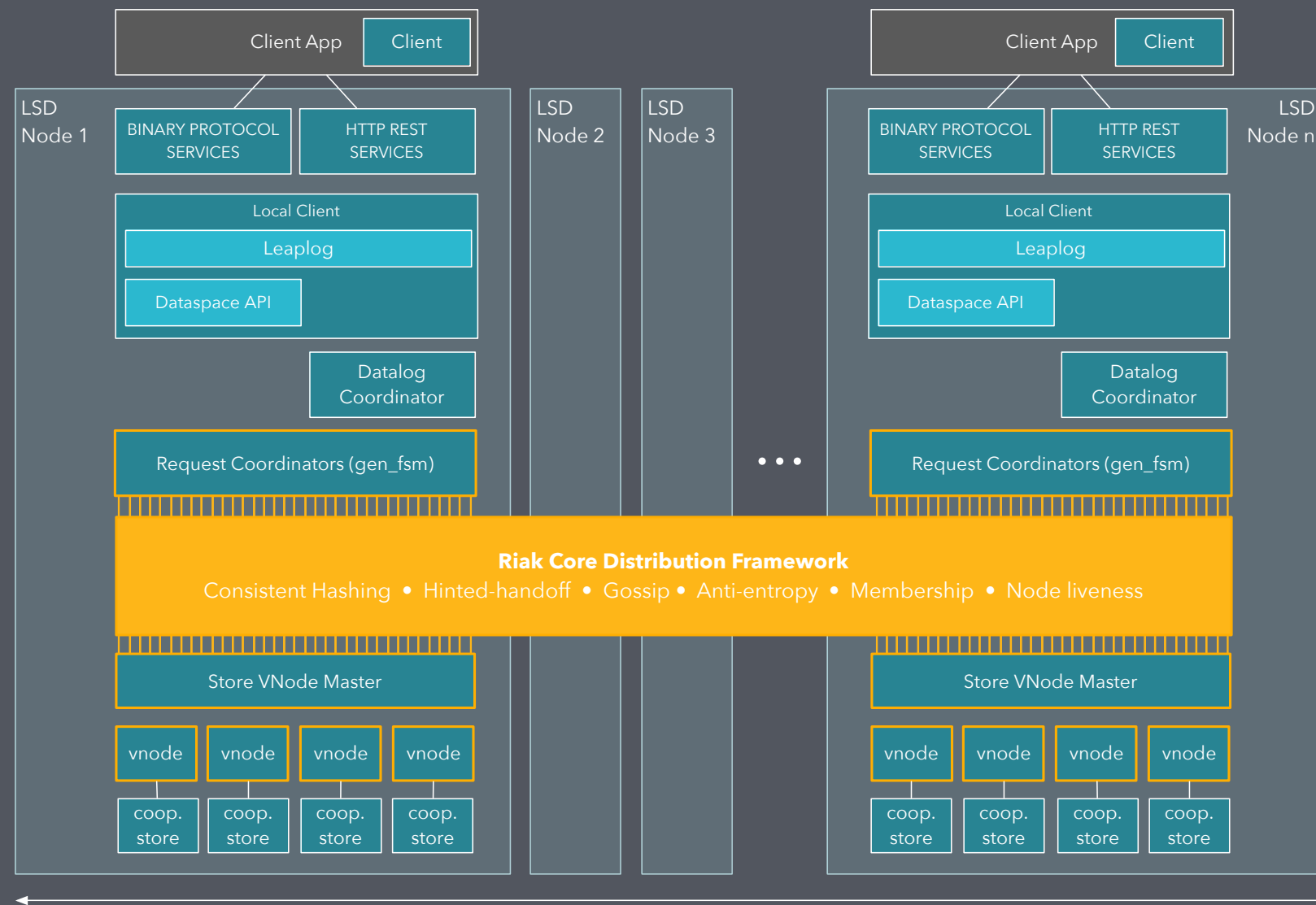
LSD

- Written in Erlang/OTP (and some C)
- Uses Basho Riak Core 2.0
- Erlang Client
- RESTfull API Services
- BERT (Binary) Services
- *JAVA client*

HOW TO USE IT

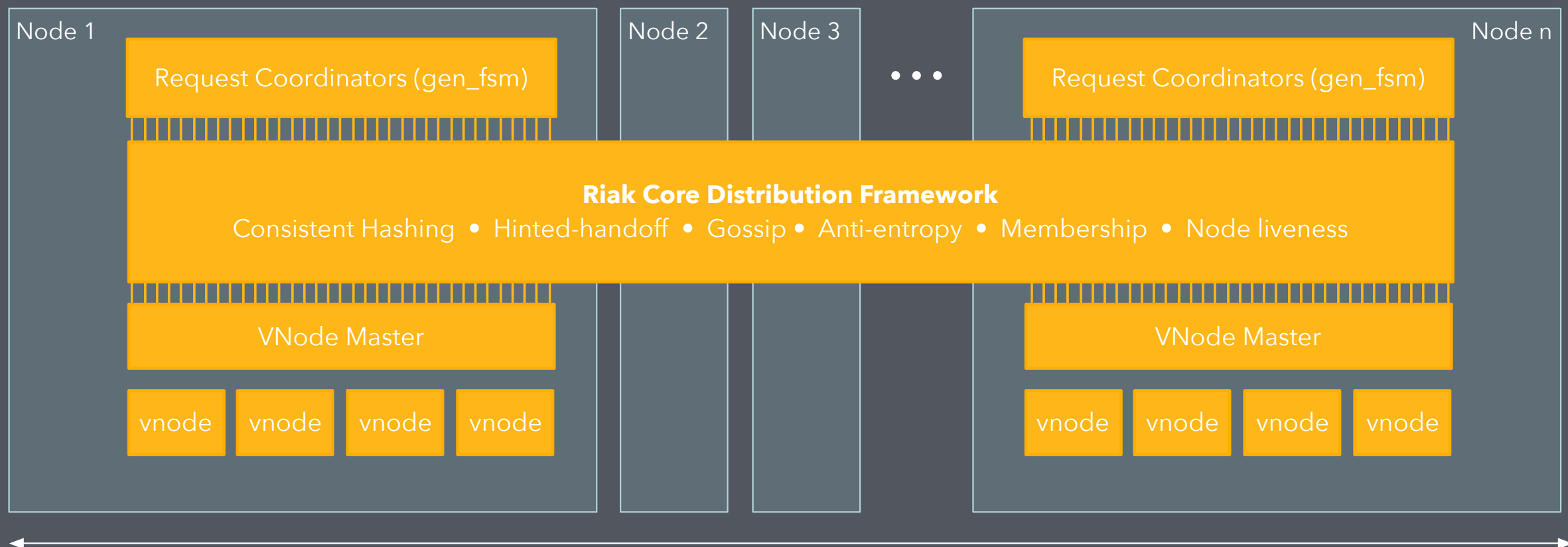


DISTRIBUTED, SCALABLE, HIGHLY-AVAILABLE, FAULT-TOLERANT



RIAK CORE²

GENERALISATION OF AMAZON'S DYNAMO ARCHITECTURE



² github.com/basho/riak_core

RIAK CORE

GENERALISATION OF AMAZON'S DYNAMO ARCHITECTURE

- Masterless, decentralised
- Homogeneous (all nodes are equal)
- Gossip protocol to sync cluster metadata / state
- Consistent Hashing for replica placement
- Active and passive anti-entropy
- Hinted Handoff (always-writable)

DISTRIBUTED WRITES

```
%% Txid is implicit
```

```
Stmt = lsd:statement(S,P,0,G,Degree)
```

```
{ok, Ctxt} = lsd:write(Client, [Stmt]).
```

```
{ok, Promise} = lsd:async_write(Client, [Stmt]).
```

```
% Remove is a write w/degree = 0
```

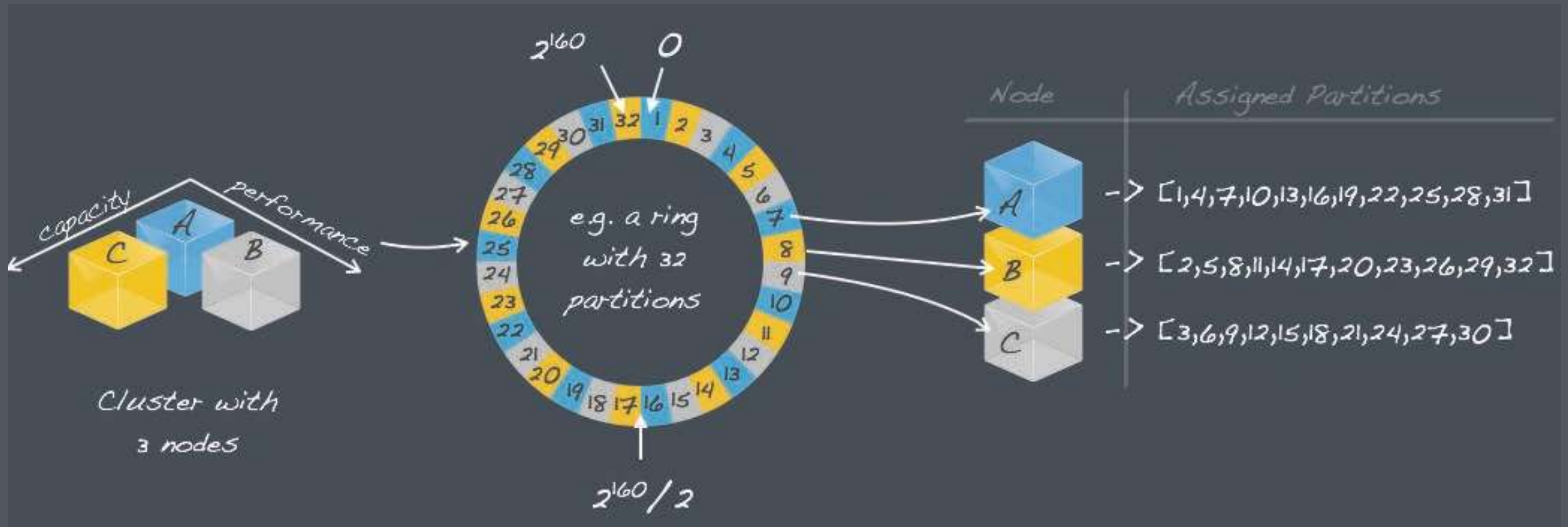
```
{ok, Ctxt} = lsd:remove(Client, [Stmt]).
```

```
{ok, Promise} = lsd:async_remove(Client, [Stmt]).
```


DISTRIBUTED WRITES

- Fully indexed: maximum of 6 covering indices with diff collation orders [GSPO, GPOS, SPOG, POGS, OGSP, OSPG]
- Index entries are distributed using Consistency Hashing to 4 Preflists (S, P, O, G)
- $n_value \leq \# \text{ vnodes} \leq 4 * n_value$
- Tunable eventual consistency idem Riak KV
(n_value , w , pw , dw)

CONSISTENT HASHING



CONSISTENT HASHING

- $\text{hash}(G) \rightarrow [\text{GSP0}, \text{GP0S}]; \text{hash}(S) \rightarrow \text{SPOG}; \dots$
- Every Node hosts a subset of the each index
- Each VNode represents a ring partition
- Data locality: all data for a given resource (in S, P, 0 or G position) is located in the same VNode

ACTIVE ANTI-ENTROPY (AAE)

- Based on Riak Core AAE
- Members of a preference list periodically exchange information
- Information -> merkle tree of txids => cheap
- Only differing tree branches are exchanged
- Disk-based Command Log is mapping of txid to command (Request)
- Healing consists in replaying commands

DISTRIBUTED READS

```
%% Degree treated as range query >=
Pattern = lsd:pattern({var,1},"<urn:profile:age>",{var,<<"Age">>},'_',0.5).

%% Sync
{ok, ResultSet} = lsd:read(Client, Pattern).

{ok, ResultSet} = lsd:take(Client, Pattern).

%% Async
{ok, Promise} = lsd:async_read(Client, [Stmnt]).

{ok, Promise} = lsd:async_take(Client, [Stmnt]).

%% Temporal Queries
lsd:read(Client, Pattern, {txid, Yesterday}).

%% Include all changes
lsd:history(Client, PatternOrStatement)
```


DISTRIBUTED READS

```
Pattern = lsd:pattern({var,1},"<urn:profile:age>",{var,<<"Age">>},'_',0.5).
```

```
C = {condition, positive, '<', [{var,<<"Age">>}, 21]}.
```

```
lsd:read(Client, graph_pattern([P], [C])).
```

DISTRIBUTED READS

- Based on Pattern Matching (Erlang 😊)
- Returns a set of Statements
- take removes the matching statements
- history includes all versions
- Every operation is performed against a temporal view of the dataspace
- Configurable eventual consistency idem Riak KV

PASIVE ANTI-ENTROPY

- Replicas (VNodes) replies includes a signature (set of all `txid` visited by query, including tombstones)
- Diff signatures => missing data
- Triggers repair based on missing `txid`
- Reuses AAE

LEAPLOG

DATALOG

- A declarative logic programming/query language
- A subset of Prolog
- Relational Algebra + recursion
- Several extensions researched for negation, arithmetic comparisons, function symbols and aggregates
- More expressive than SQL when including negation and arithmetic comparisons

LEAPLOG

- A declarative logic programming/query language based on Datalog Adapted to work with RDF
- LSD can make deductions based on rules expressed in Leaplog Top-down, set-oriented evaluation (QSQ)
- Built-in predicates (boolean and arithmetic comparisons)
- Function symbols (built-in functions)

LEAPLOG SYNTAX

```
@prefix erl : <http://www.erlang.org#>.
```

```
(S1, erl:colleague, S2, G) :-    (S1, erl:creator, X, G),  
                                (S2, erl:creator, X, G),  
                                S1 /= S2.
```

```
(S2, erl:colleague, S1, G) :-    (S1, erl:colleague, S2, G),  
                                S1 /= S2.
```

```
(S1, erl:knows, S2, G) :-    (S1, erl:colleague, S2, G),  
                             S1 /= S2.
```

```
(S2, erl:knows, S1, G) :-    (S1, erl:knows, S2, G),  
                             S1 /= S2.
```

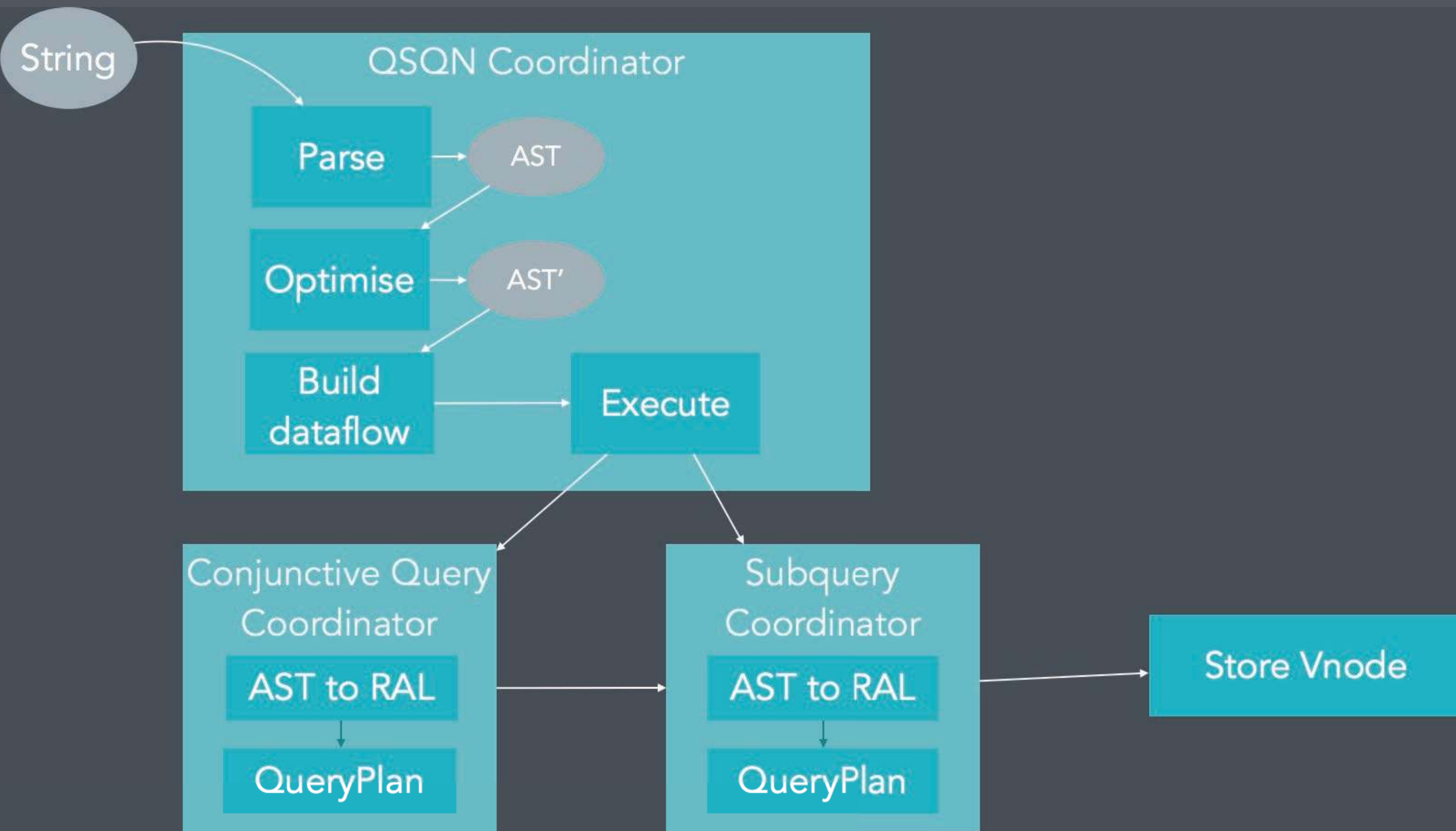
```
%% Who are colleagues  
?(S, erl:colleague, 0, <lsd:codemesh:demo>).
```

```
%% Who knows who?  
?(S, erl:knows, 0, <lsd:codemesh:demo>).
```

```
%% Everything we know about Joe  
?(erl:joe_armstrong, P, 0, <lsd:codemesh:demo>).
```

```
%% Everything we know about everybody  
?(S, P, 0, <lsd:codemesh:demo>).
```

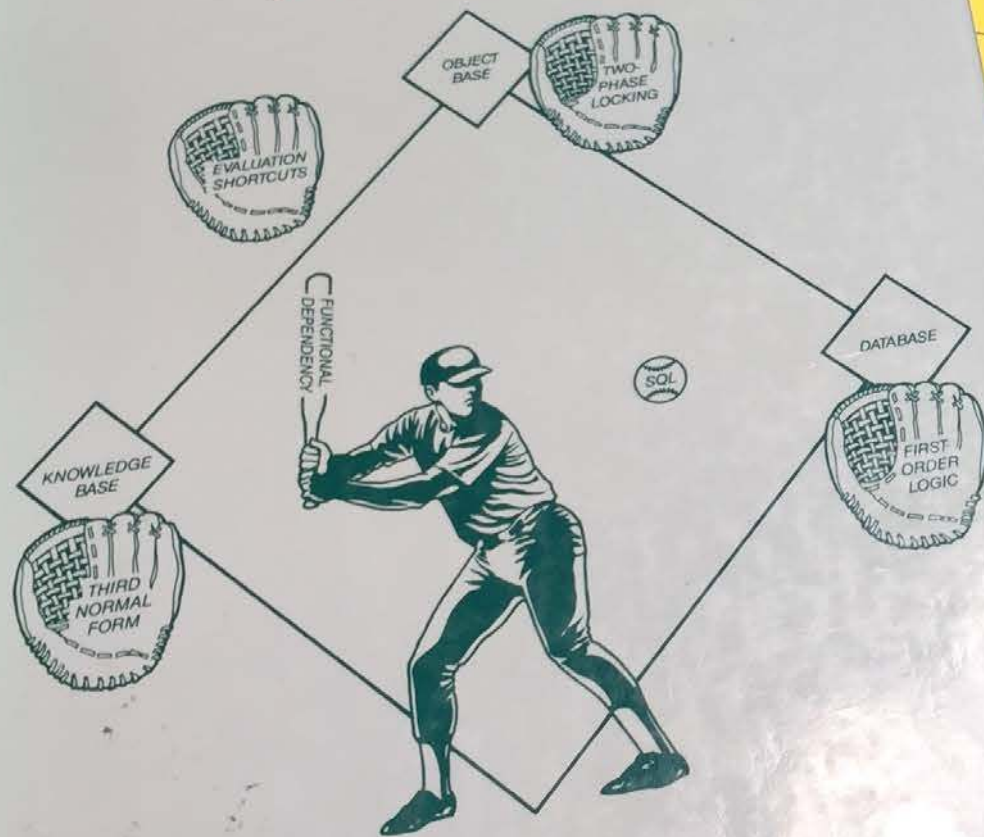
DEMO



PRINCIPLES OF DATABASE AND KNOWLEDGE-BASE SYSTEMS

VOLUME I: CLASSICAL DATABASE SYSTEMS

Jeffrey D. Ullman

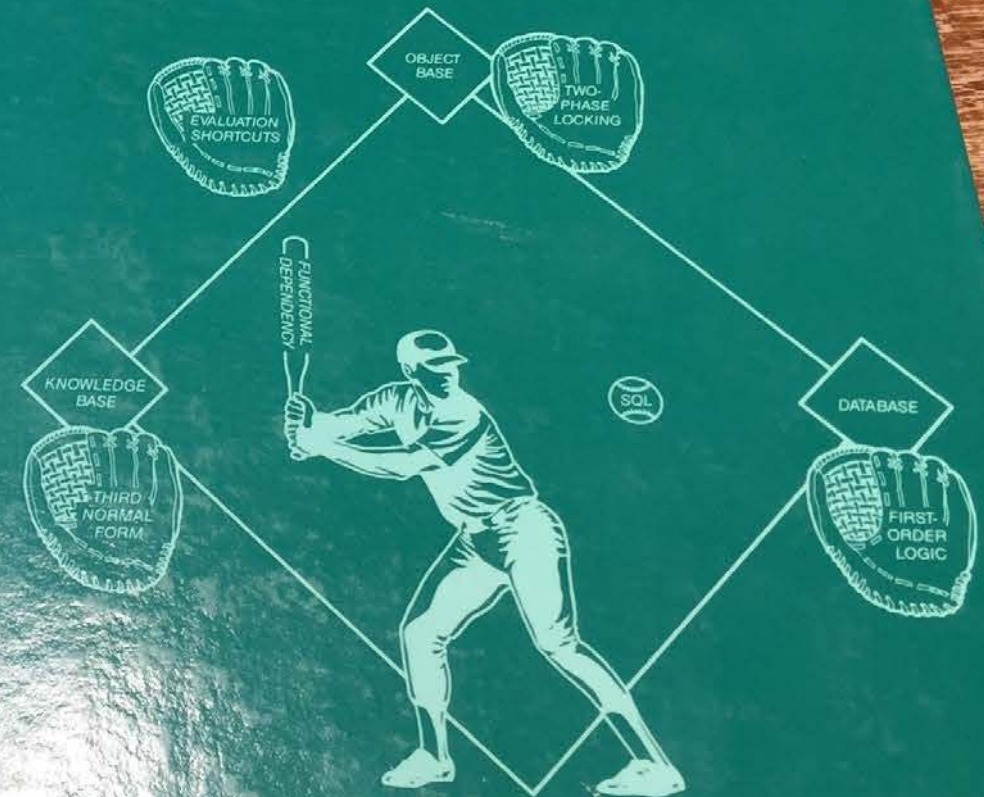


COMPUTER SCIENCE PRESS

PRINCIPLES OF DATABASE AND KNOWLEDGE-BASE SYSTEMS

VOLUME II: The New Technologies

Jeffrey D. Ullman



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HOW TO GET IT

- Not OSS (yet)
- Working on a free tier (talk to us)
- Private Beta
- @leapsight
- @aramallo

QUESTIONS?